

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of M.Sc. (Mathematics) Examination March 2018
MA712 Functions of Several Real Variables

Date: 27/03/2018, Tuesday Time: 10:00 a.m. to 10:45 a.m. Maximum Marks: 20

MCQ

Important Instructions:

1. Tick the correct answer only in this sheet.
 2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 3. Maximum time to answer Multiple Choice Questions will be 45 minutes.
 4. Each question carries one mark.
 5. In this question paper $\mathbb{R}$ denotes the field / space of all real numbers.
 6. V and W denote finite dimensional vector spaces over $\mathbb{R}$.
 7. Notations and terminologies are standard.
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Q - I Choose the correct answer from the given options in the following:

1. Define $f : \mathbb{R}^2 \mapsto \mathbb{R}$ as $f(x, y) = \cos(x + y)$. Then $O(f, (\frac{1}{2}, \frac{1}{2})) = \underline{\hspace{2cm}}$.
 (a) $\frac{1}{2}$ (b) 2 (c) 1 (d) 0
2. Which of the following is not true for a linear map $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
 (a) $Df(e_4)(e_5) = 1$ (c) $Df(e_4)$ is differentiable
 (b) f is differentiable (d) f is continuous
3. Which of the following is true for $x, y \in \mathbb{R}^n$?
 (a) $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ (c) $\|x\| \leq \sum_{i=1}^n |x_i|$
 (b) $|\langle x, y \rangle| = \|x\| \|y\|$ (d) $\|x\| + \|y\| \leq \|x - y\|$
4. Define $f : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ as $f(x, y, z) = (0, y + z, 0, z - x)$. Then, Which of the following is true ?
 (a) f is differentiable at $(0, 0, 0)$ and $Df(0, 0, 0) = 2f$
 (b) f is differentiable at $(0, 0, 0)$ and $Df(0, 0, 0) = f$
 (c) f is not continuous at $(0, 0, 0)$
 (d) f is not differentiable at $(0, 0, 0)$
5. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $D_j f(a)$ exists for all $1 \leq j \leq n$. Then _____.
 (a) f is continuous at a
 (b) f is continuously differentiable at a
 (c) f is differentiable at a
 (d) None of these
6. Let $T, S \in \mathcal{T}^2(\mathbb{R}^2)$ be defined as $T(x, y) = x_1 y_1 - x_2 y_2$ and $S(x, y) = x_1 y_2$, where $x = (x_1, x_2)$ $y = (y_1, y_2)$. Then $\text{Alt}(T \otimes S) = \underline{\hspace{2cm}}$.
 (a) $S \otimes T$ (b) $T \otimes S$ (c) $T \wedge S$ (d) 0
7. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $|f(x)| \leq \|x\|^{2018}$. Then _____.
 (a) f is not differentiable at origin
 (b) f is differentiable at origin
 (c) f is not continuous at origin
 (d) None of these
8. For $x \in \mathbb{R}^5$, define $\varphi(x) = x_4$ and $\eta(x) = x_5$. Then _____.
 (a) $\varphi \otimes \eta \in \Lambda^2(\mathbb{R}^5)$
 (b) $\varphi \otimes \eta \in \mathcal{T}^2(\mathbb{R}^5)$ but $\varphi \otimes \eta \notin \Lambda^2(\mathbb{R}^5)$
 (c) $\text{Alt}(\varphi \otimes \eta) = \varphi \otimes \eta$
 (d) $\text{Alt}(\varphi \otimes \eta) = \eta \otimes \varphi$
9. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous at $a \in \mathbb{R}^n$. Then _____.
 (a) $Df(a)$ exists (c) $D_j f(a)$ exists ($1 \leq j \leq n$)
 (b) $D_x f(a)$ exists ($x \in \mathbb{R}^n$) (d) None of these

10. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $D_x f(a)$ exists for all $x \in \mathbb{R}^n$. Then, Which of the following is true ?
- (a) f is continuous at a
 - (b) f is continuously differentiable at a
 - (c) f is differentiable at a
 - (d) None of these
11. Which of the following is true?
- (a) Every exact form is closed form
 - (b) Every closed form is exact form
 - (c) If ω is a exact form then $d(\omega) \neq 0$
 - (d) None of these.
12. Which of the following is not true for a differentiable function $f : \mathbb{R}^2 \mapsto \mathbb{R}$?
- (a) $D_1 f(a) = D_2 f(a)$, ($a \in \mathbb{R}^n$)
 - (b) $D_1 f(a)$ and $D_2 f(a)$ exist, ($a \in \mathbb{R}^n$)
 - (c) $D_x(a)$ exists, ($x, a \in \mathbb{R}^n$)
 - (d) f is continuous at a
13. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function at $a \in \mathbb{R}^n$. Then ____.
- (a) $Df(a) = (D_1 f^1(a), \dots, D_n f^m(a))$
 - (b) $Df(a) = (f^1(a), \dots, f^m(a))$
 - (c) $Df(a) = (Df^1(a), \dots, Df^m(a))$
 - (d) None of these
14. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined as $f(x, y, z) = \exp(x) + z$. Then $Df =$ ____.
- (a) $\exp(x)d\pi_2 + d\pi_3$
 - (b) $d\pi_1 + \exp(x)d\pi_3$
 - (c) $\exp(x)d\pi_1 + d\pi_3$
 - (d) $d\pi_2 + \exp(x)d\pi_3$
15. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in \mathbb{R}^n$. Then ____.
- (a) $Df(a)$ is not continuous at a
 - (b) $Df(a)$ is differentiable at a
 - (c) $Df(a)$ is continuously differentiable at a
 - (d) $Df(a)$ is a constant function.
16. Which of the following is true for $T, S \in \mathcal{T}^3(\mathbb{R})$?
- (a) $T(2018, 2017, 0) \neq S(0, 2017, 2018)$
 - (b) $T(0, 2017, 2018) = S(2018, 2017, 0)$
 - (c) $T(2018, 0, 2017) \neq S(2018, 0, 2017)$
 - (d) $T(0, 0, 2018) \neq S(0, 0, 2018)$
17. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in \mathbb{R}^n$ and let $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$ be a differentiable at $f(a)$. Then ____.
- (a) $D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$
 - (b) $D(g \circ f)(a) = Df(a) \circ Dg(f(a))$
 - (c) $D(f \circ g)(a) = Dg(f(a)) \circ Df(a)$
 - (d) None of these

18. Let $f : V \rightarrow W$ be a linear map and let $S \in \mathcal{T}^k(W)$, $T \in \mathcal{T}^\ell(W)$. Then ____.
- (a) $f_{k+\ell}^*(S \otimes T) = f_\ell^*(S) \otimes f_k^*(T)$ (c) $f_{k+\ell}^*(S \otimes T) = f_k^*(S) \wedge f_\ell^*(T)$
(b) $f_{k+\ell}^*(S \otimes T) = f_k^*(S) \otimes f_\ell^*(T)$ (d) $f_{k+\ell}^*(S \otimes T) = f_\ell^*(S) \wedge f_k^*(T)$
19. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x) = \sin(x_1)$, where $x = (x_1, x_2)$. Then for each $a \in \mathbb{R}^2$, $Df(a)(x) = \underline{\hspace{2cm}}$.
- (a) $\cos(a_2)x_2$ (c) $\cos(a_1)x_1$
(b) $\cos(a_1 + a_2)x_2$ (d) $\cos(a_1 + a_2)x_1$
20. Which of the following is not true for $S \in \mathcal{T}^k(V)$ and $T \in \mathcal{T}^\ell(V)$?
- (a) $S \otimes T \in \mathcal{T}^{k+\ell}(V)$ (c) $(\alpha S) \otimes T = S \otimes (\alpha T)$, $(\alpha \in \mathbb{R})$
(b) $S \otimes T = T \otimes S$ (d) $T \otimes S \in \mathcal{T}^{\ell+k}(V)$
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MA712 Functions of Several Real Variables

Date: 27/03/2018, Tuesday Time: 10:45 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

1. **Section I and Section II must be written in separate answer books.**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. In this question paper, $\mathbb{R}$ denotes the field / space of all real numbers.
5. V and W denote finite dimensional vector spaces over $\mathbb{R}$.
6. Notations and terminologies are standard.

SECTION I

Q - II (A) Attempt any TWO of the following: **[08]**

- (1) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in \mathbb{R}^n$. Show that there exists a unique linear map $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - \lambda(h)\|}{\|h\|} = 0.$$

- (2) Prove that each inner product T on a vector space V has an orthonormal basis with respect to T .
- (3) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable and let $k \in \mathbb{N}$. Prove that the map $\widetilde{f_{k*}} : \Delta_{kF}(\mathbb{R}^m) \rightarrow \Delta_{kF}(\mathbb{R}^n)$ is a linear map.

(B) Attempt any SIX of the following: **[12]**

- (1) For $x, y \in \mathbb{R}^2$, define $T : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ as

$$T(x, y) = \frac{1}{2}(x_2y_1 - x_1y_2), \text{ where } x = (x_1, x_2), y = (y_1, y_2).$$

Show that $T \in \Lambda^2(\mathbb{R}^2)$.

- (2) Define a vector field and a k-form on $\mathbb{R}^n$.
- (3) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. Then define df and show that df is a 1-form on $\mathbb{R}^n$.
- (4) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined as $f(x, y) = (y, x)$. Show that for any $P \in \mathbb{R}^2$, $\widetilde{f_{*1}}(\pi_{1P}) = \pi_{2P}$.
- (5) Define $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ as $f(x, y, z) = (x - y, y - z, x - z)$. Find the matrix A such that $f(x, y, z) = (x \ y \ z)A$, $(x, y, z) \in \mathbb{R}^3$.

- (6) Prove that a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous.
- (7) Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ as $f(x, y) = |xy|$. Show that f is differentiable at origin.
- (8) Let T be a k -tensor on vector space V . Show that $\text{Alt}(T) \in \Lambda^k(V)$.
- (9) Let $\varphi : V \rightarrow \mathbb{R}$ be a linear map. Show that $\varphi \wedge \varphi = 0$.

SECTION II

Q - III (A) Attempt any THREE of the following: [18]

- (1) Let $A \subset \mathbb{R}^n$ and $f : A \rightarrow \mathbb{R}$ be a bounded function. Then show that f is continuous at $a \in A$ if and only if $O(f, a) = 0$.
- (2) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $a \in \mathbb{R}^n$. Then prove that f is differentiable at a if and only if each component $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$ ($1 \leq i \leq m$) is differentiable at a .
- (3) Let V be a vector space of dimension n . Show that the dimension of the vector space $\mathcal{T}^k(V)$ is n^k .
- (4) Let $x, y \in \mathbb{R}^n$. Then prove that $|\langle x, y \rangle| \leq \|x\| \|y\|$.
- (5) Let ω be k -form and η be a ℓ -form. Then prove the following.
 1. $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^k \omega \wedge d(\eta)$
 2. $d(d\omega) = 0$.

(B) Attempt any THREE of the following: [12]

- (1) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq \|x\|^2$. Show that f is differentiable at origin.
- (2) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x, a \in \mathbb{R}^n$. Show that if f is differentiable at a then $D_x f(a) = Df(a)x$.
- (3) Let ω be an alternating k -tensor on vector space V . Show that $\text{Alt}(\omega) = \omega$.
- (4) Let $\varphi, \eta : \mathbb{R}^5 \rightarrow \mathbb{R}$ be defined as $\varphi(x) = x_1 + x_2$ and $\eta(x) = x_4 + x_5$, where $x = (x_1, x_2, x_4, x_5)$. Does $\varphi \otimes \eta \in \Lambda^2(\mathbb{R}^5)$? Also find $\text{Alt}(\varphi \otimes \eta)$.
- (5) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. Show that $df = D_1 f d\pi_1 + \dots + D_n f d\pi_n$.